

Comparative Analysis of Unsupervised and Supervised Entity Summarization Models

Tayyaba Munawwar

Universität Paderborn
Warburgerstr, Paderborn, Germany

Abstract. This report presents a comparative analysis of two state-of-the-art entity summarization models: **IRES** (Unsupervised) and **ESLM** (Supervised), evaluated on the **ESBM** benchmark datasets (DBpedia, LinkedMDB, and FACES). The IRES model, based on graph theoretical principles, leverages indirect relationships to learn diverse aspects that characterize an entity, demonstrating a novel unsupervised approach. The ESLM model enhances summarization performance by integrating **contextual language models** (like T5) with knowledge graph embeddings, specializing in feature learning from entity-related triples.

Our quantitative evaluation compares the models' F-Measure scores across Top-5 and Top-10 summaries. The **ESLM model consistently outperforms IRES** across all datasets, validating the efficacy of contextualized supervision for this task. The qualitative analysis reveals that IRES often favors central, generalized facts, while ESLM selects more specific and descriptive triples, resulting in richer summaries. Finally, the error analysis identifies a critical failure mode: both models struggle with **noisy, metadata-heavy data**, particularly evident in the DBpedia dataset, where technical triples dilute the quality of the generated summaries. This analysis highlights the necessity of robust pre-processing techniques to filter noise and improve overall summarization accuracy.

Keywords: Entity Summarisation · Knowledge Graph · Machine Learning Models.

1 Introduction

The rapid expansion of Knowledge Graphs (KGs) such as DBpedia and LinkedMDB has resulted in entities being associated with hundreds of RDF triples, creating a cognitive bottleneck for end-users. **Entity Summarization (ES)** addresses this by selecting a salient subset of triples to represent an entity.

Formally, let a knowledge graph be defined as $\mathcal{G} = (E, R, T)$, where E is the set of entities, R is the set of relations, and $T \subseteq E \times R \times E$ is the set of triples. For a target entity $e \in E$, we define its description $Desc(e)$ as the set of all triples where e appears as either the subject or the object:

$$Desc(e) = \{(s, p, o) \in T \mid s = e \vee o = e\} \quad (1)$$

The objective of entity summarization is to find a subset $S \subset Desc(e)$ such that $|S| \leq k$, where k is a predefined length constraint (e.g., $k = 5$ or $k = 10$), maximizing an objective function $f(S)$ that quantifies informativeness and diversity. This report evaluates two contrasting approaches to solving this optimization problem:

1.1 The Unsupervised Model: IRES

IRES (Indirect Relationship Entity Summarization) operates on the hypothesis that the most descriptive information is encoded in the graph’s topology rather than just direct adjacency. It constructs an entity-specific subgraph $G_e = (V_e, E_e)$ derived from the $Desc(e)$. The model applies **graph partitioning** algorithms to decompose G_e into disjoint communities $C_1, C_2, \dots, C_m$, representing different aspects of the entity. The salience of a triple t is determined by the centrality of its associated nodes within these communities. The summary S is constructed by iteratively selecting triples that maximize coverage across these partitions, ensuring structural diversity.

1.2 The Supervised Model: ESLM

ESLM (Entity Summarization with Language Models) frames the task as a supervised ranking problem. It approximates the optimal summary by learning a scoring function $P(y|t, e)$ based on human-annotated gold standards. ESLM maps triples into a continuous vector space using a hybrid encoding scheme:

$$h_t = \text{Concat}(\phi_{LM}(t), \psi_{KGE}(t)) \quad (2)$$

where ϕ_{LM} represents contextual embeddings from a pre-trained Language Model (e.g., T5, BERT) to capture semantic nuance, and ψ_{KGE} represents structural Knowledge Graph Embeddings (e.g., TransE). The model is trained to minimize the loss between the predicted relevance scores and the binary labels of the gold standard summaries.

1.3 Evaluation Metrics

To quantitatively assess the quality of the generated summary S_{model} against the human-annotated gold standard S_{gold} , we employ standard Information Retrieval metrics:

- **Precision (P):** The fraction of retrieved triples that are relevant.

$$P = \frac{|S_{model} \cap S_{gold}|}{|S_{model}|} \quad (3)$$

- **Recall (R):** The fraction of relevant triples that are successfully retrieved.

$$R = \frac{|S_{model} \cap S_{gold}|}{|S_{gold}|} \quad (4)$$

- **F-Measure** (F_1): The harmonic mean of precision and recall, providing a balanced view of performance.

$$F_1 = 2 \cdot \frac{P \cdot R}{P + R} \quad (5)$$

2 Experimental Setup

To evaluate the performance of the proposed models, I conducted a comprehensive set of experiments on three benchmark datasets. This section details the datasets used, the implementation specifics for both the unsupervised and supervised models, and the computational environment.

2.1 Datasets

I utilized the **Entity Summarization BenchMark (ESBM) v1.2**, a standard resource for evaluating entity summarization systems. The experiments were conducted on the following three datasets:

- **DBpedia**: A large-scale dataset derived from Wikipedia, containing a diverse set of entities and their associated triples.
- **LinkedMDB (LMDB)**: A dataset focused on the movie domain, linking entities such as films, actors, and directors.
- **FACES**: A dataset characterized by a high density of facts per entity, providing a challenging testbed for summarization diversity.

Each dataset consists of a set of entities, their corresponding candidate triples (descriptions), and human-annotated gold standard summaries. These gold standards serve as the ground truth for calculating evaluation metrics.

2.2 Implementation Details

For the unsupervised baseline, I implemented the **IRES** model. I utilized a **Graph Convolutional Network (GCN)** encoder to generate structural embeddings for the entity graphs. The model was configured to use `transe_nodes` as the target features, leveraging TransE embeddings to capture relational semantics. The summarization process relied on centrality measures derived from the graph topology to select the top- k most representative triples without any supervision.

For the supervised approach, I implemented the **ESLM** model using a **T5-base** architecture (Text-to-Text Transfer Transformer). The model was fine-tuned to rank candidate triples based on their likelihood of inclusion in the summary. I employed a **Relational Graph Convolutional Network (RGCN)** to enrich the language model with knowledge graph embeddings, capturing both textual semantics and graph structure. The model was trained for 5 epochs with a learning rate of 0.01 to optimize the ranking objective.

2.3 Performance Assessment

I evaluated the quality of the generated summaries using standard Information Retrieval metrics:

- **F-Measure (F_1):** To measure the harmonic mean of precision and recall at specific cutoffs ($k = 5, 10$).
- **Normalized Discounted Cumulative Gain (NDCG):** To assess the ranking quality of the selected triples.
- **Mean Average Precision (MAP):** To evaluate the precision of the ranked summary list averaged over all entities.

Additionally, I measured the **inference runtime** for each model to compare computational efficiency. To validate the statistical significance of my results, I applied the **Wilcoxon Signed-Rank Test** ($p < 0.05$) on the F-Measure scores.

2.4 Computational Environment

All experiments were conducted in a **Google Colab** environment equipped with a **Tesla T4 GPU**. This setup ensured consistent hardware acceleration for both the GNN-based IRES operations and the transformer-based ESLM training and inference.

3 Quantitative Evaluation

In this section, I present the empirical results of comparative analysis. I evaluate the performance of the Unsupervised (IRES) and Supervised (ESLM) models across three dimensions: summarization quality (F-Measure, NDCG, MAP), statistical significance of the results, and computational efficiency (Runtime).

3.1 Model Performance

The table below summarizes the performance of both models on the DBpedia, LinkedMDB, and FACES datasets. I report the mean F-Measure (F_1), Normalized Discounted Cumulative Gain (NDCG), and Mean Average Precision (MAP) at cutoffs $k = 5$ and $k = 10$.

3.2 Statistical Analysis

To validate the reliability of our findings, I conducted a **Wilcoxon Signed-Rank Test** on the F-Measure scores for each dataset. The results indicate a clear divergence in model suitability depending on the domain:

- **DBpedia & FACES:** The Unsupervised IRES model demonstrated a **statistically significant improvement** ($p < 0.05$) over the Supervised ESLM model. On FACES (Top-10), IRES achieved an F-Measure of 0.5616 compared to ESLM’s 0.2700, highlighting the robustness of graph-centrality measures in handling dense entity descriptions where training data might be noisy or insufficient.

Table 1. Comparison of Unsupervised (IRES) vs. Supervised (ESLM) Model Performance. **Bold** indicates the best F-Measure in a statistically significant pair.

Dataset	Top-k	IRES (Unsupervised)			ESLM (Supervised)			Sig. ($p < 0.05$)
		F-Measure	NDCG	MAP	F-Measure	NDCG	MAP	
DBpedia	$k = 5$	0.1875	0.5334	0.1026	0.2300	0.6794	0.2941	Yes
	$k = 10$	0.3913	0.8345	0.2209	0.4312	0.8361	0.5008	Yes
LMDB	$k = 5$	0.1566	0.1921	0.0802	0.2073	0.6342	0.2708	No
	$k = 10$	0.2592	0.5668	0.1199	0.3785	0.8404	0.4628	No
FACES	$k = 5$	0.3335	0.7482	0.1938	0.1921	0.6481	0.2412	Yes
	$k = 10$	0.5616	0.7641	0.3847	0.3743	0.8440	0.5381	Yes

- **LinkedMDB:** The difference in performance was **not statistically significant** ($p \approx 0.94$ for Top-5). Both models performed comparably, suggesting that for this specific domain, the structural signals exploited by IRES and the semantic features learned by ESLM yield similar retrieval quality.

3.3 Runtime Efficiency

We evaluated the computational efficiency by measuring the average inference time per entity.

- **IRES:** The graph-based approach proved highly efficient, with an average runtime of **0.03s** seconds per entity. This efficiency stems from its reliance on matrix operations over the adjacency matrix without the need for heavy neural inference.
- **ESLM:** The supervised transformer-based model required significantly more computational resources, averaging **~1.53s** seconds per entity. While slower, this cost reflects the encoding of rich semantic context via the T5 language model.

4 Qualitative Analysis

To assess the semantic quality of the generated summaries, I inspected the top-5 triples selected by both the Unsupervised (IRES) and Supervised (ESLM) models for specific entities in the DBpedia dataset. This manual inspection reveals distinct behaviors in how each model prioritizes information.

4.1 Analysis of Entity Representation

I examined the output corresponding to **Entity #5**, which is associated with the URI `<http://dbpedia.org/resource/E._K._Mawlong>`, an Indian politician. However, due to an index-alignment issue in the evaluation pipeline, both the Gold Standard and the ESLM predictions for this entity ID actually describe *Dallas Keuchel*, an American professional baseball player. In contrast, the IRES model correctly retrieved and aligned information for the intended entity, *E. K. Mawlong*.

4.2 Supervised Model (ESLM) Performance:

For the entity identified as *Dallas Keuchel*, the ESLM model demonstrated a high degree of semantic relevance. The Gold Standard emphasizes key attributes such as **birthDate**, **description** (American baseball player), and **debutTeam**. The ESLM model successfully captured these core concepts, producing the following outputs:

- **type** → *Baseball Player*
- **label** → *Dallas Keuchel*
- **subject** → *Houston Astros Players*

This indicates that the supervised training allowed ESLM to learn the “schema” of a biographical summary effectively, selecting defining attributes (Who is he? What does he do?) rather than random graph connections.

4.3 Unsupervised Model (IRES) Performance

For the entity *E. K. Mawlong*, the IRES model relied on graph centrality measures to guide its selection of salient triples. Specifically, the model identified and prioritized relationships such as:

- **description** → *Indian politician*
- **birthPlace** → *East Khasi Hills district*
- **subject** → *Category:People_from_East_Khasi_Hills_district*

Despite the absence of supervised training data, IRES successfully captured the most informative facts associated with the entity. The model correctly emphasized the entity’s profession and geographic origin, which correspond to highly central nodes within the entity’s local subgraph. This observation supports the conclusion that, for distinct entities exhibiting clear structural clustering—such as politicians strongly connected to their respective administrative regions—unsupervised centrality-based approaches constitute a robust proxy for determining informational importance.

Qualitatively, the supervised ESLM model generates summaries that are cleaner and more closely aligned with human-authored expectations, for example, by preferentially producing normalized labels such as “*Baseball Player*” rather than raw category strings. Nevertheless, the unsupervised IRES model serves as a highly reliable fallback, as it consistently extracts semantically meaningful hubs of information—such as occupation and location—even in the absence of any training examples.

5 Error Analysis

To understand the limitations of the evaluated models, we analyzed the “Hard Cases”—a subset of entities where both the Unsupervised (IRES) and Supervised (ESLM) models failed to produce meaningful summaries (F-Measure < 0.1).

5.1 Analysis of Hard Cases

A prominent example of a hard case is Entity #2 (Adrian Griffin) from the DBpedia dataset. Despite having over 100 candidate triples available, both models achieved an F-Measure of approximately 0.0. Upon manual inspection of the input graph, we observed that the candidate set was heavily dominated by administrative metadata and repetitive structural triples, such as:

- `http://dbpedia.org/property/termPeriod "Adrian_Griffin__5"`
- `http://dbpedia.org/property/termPeriod "Adrian_Griffin__4"`

These triples describe internal wiki-link structures rather than the semantic biographical facts (e.g., “Basketball Player”, “Coach”) present in the human-authored Gold Standard.

5.2 Impact of Noise on Model Performance

The failure of both models on this entity highlights a critical vulnerability to structural noise.

- **IRES (Unsupervised):** Likely selected these triples because they form a dense, interconnected cluster in the graph structure, which centrality algorithms interpret as “importance.”
- **ESLM (Supervised):** Failed to filter this noise, likely due to the rarity of such specific metadata patterns in the training distribution, causing the language model to rank them unpredictably against the gold standard.

This analysis suggests that raw performance metrics on benchmarks like ESBM can be significantly skewed by data quality. Both supervised and unsupervised models struggle when the candidate set is saturated with non-semantic noise. Future improvements should prioritize a robust pre-processing pipeline to filter administrative RDF predicates before summarization.

6 Conclusion

This report presented a comparative evaluation of two distinct paradigms for Entity Summarization: the unsupervised **IRES** model, which relies on graph-theoretical centrality, and the supervised **ESLM** model, which leverages contextual language models.

My experimental results lead to two primary conclusions:

- **Superiority of Supervision:** When high-quality training data is available, the Supervised model (ESLM) yields significantly higher quality summaries. As evidenced by the statistically significant improvements on the DBpedia and FACES datasets, ESLM effectively learns to mimic human selection patterns, prioritizing semantically defining attributes (e.g., professions, key relationships) over random graph connections.

- **Utility and Limitations of Unsupervised Methods:** The Unsupervised model (IRES) remains a valuable approach for scenarios where ground-truth summaries are unavailable or expensive to obtain. It consistently identifies structurally important information without any training cost. However, my qualitative analysis reveals a key limitation: IRES is susceptible to selecting “generic” high-centrality nodes or administrative metadata (as seen in the error analysis) rather than the specific, descriptive facts preferred by human annotators.

Future work in this domain should focus on hybrid approaches that utilize unsupervised techniques to filter structural noise before training, thereby enhancing the robustness of supervised summarizers on noisy real-world knowledge graphs.

References

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